



Klausur Sommersemester 2021, Fragen

Kognitive Systeme (Technische Universität München)



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Registration number

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Note:

- Cross your Registration number(with leading zero). It will be evaluated automatically.
- Sign in the corresponding signature field.

Cognitive Systems

Exam: IN2222 / Endterm

Date: Friday 6th August, 2021

Examiner: Prof. Dr. Alois Knoll

Time: 08:00 – 09:15

	P 1	P 2	P 3	P 4	P 5	P 6
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Working instructions

- This exam consists of **12 pages** with a total of **6 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 120 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources: None
- Write your name, seat number and lecture hall below the signature field on the first page.
- Write your name and matriculation number on the top of **every** page.
- Give **concise** answers using appropriate **terminology**.
- You may answer in English and German. Do not mix languages within a question.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag, and close the bag.
- You have **75 minutes** working time.

Left room from _____ to _____ / Early submission at _____

Problem 1 Cognition (18 credits)

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1.1 Briefly explain the difference between cognition and intelligence.

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1.2 Should robots always be built more human-like to reach better acceptance with human users? Name the underlying phenomenon and briefly explain its outcome.

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1.3 Explain the difference between a bio-inspired model and a purely computational model as defined in the lecture.

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1.4 List and explain two main properties of Dynamical Systems.

Problem 2 Neurorobotics platform and cognitive architectures (18 credits)

2.1 Name three types of Cognitive Architectures.

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2.2 Explain how ACT-R implements parallel information processing and how it implements serial information processing.

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2.3 Describe the buffer test and buffer action of the following production rule using ACT-R terminology.

```
(p type
=goal>
isa goal
state enter-number
==>
+MANUAL>
CMD PRESS-KEY
KEY "2"
)
```

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2.4 Name three advantages of experimenting with simulated robots instead of physical robots.

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2.5 Explain the purpose of the MapSpikeSource and the MapSpikeSink devices of a Transfer Function in the Neurorobotics Platform.

Problem 3 Human Brain (13 credits)

3.1 Provide the names of the three main components of a biological neuron.

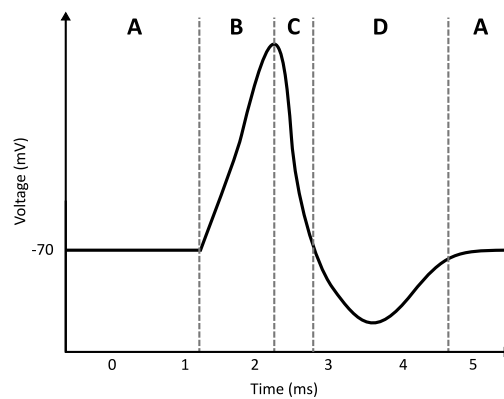
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3.2 Explain whether it is true that only ten percent of the brain is used.

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3.3 Assign the correct names to the different phases **A**, **B**, **C**, and **D** of the action potential depicted in the figure below.

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3.4 Name the two streams of visual processing in the human brain and briefly outline the type of information computed along each of the streams.

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Problem 4 Biological neural networks (29 credits)

The membrane potential of a simplified Leaky Integrate and Fire (LIF) neuron is described as a single differential equation

$$C \frac{dv}{dt} = I(t) - \frac{v(t)}{R}, \quad (4.1)$$

with membrane capacity C , membrane resistance R , and input current $I(t)$. Spikes of LIF neurons are generated if the membrane potential reaches a threshold value v_{thresh} . After the occurrence of a spike the membrane potential is immediately set to $v_{\text{reset}} = v(0)$, i.e., we have no refractory period.

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4.1 Name two essential features of neural processing that are captured by the *leaky integrate-and-fire* model.

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4.2 Solve the differential equation 4.1 for the case of a constant input current $I(t) = I$, given an initial membrane potential $v(0) = 0$ at initial time $t_0 = 0$.

4.3 State the two main schemes of encoding techniques for spiking neural networks. Name the main advantage for each of them.

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4.4 State the basic principle of Hebbian learning.

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4.5 In the context of Hebbian learning, describe what "saturation" means, and what is the problem when a Hebbian learning algorithm does not possess this feature. You might draw a graph for illustrating the problem.

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4.6 Equation (4.6) is a basic form of Hebbian learning. Modify the equation (or add a second one) in order to add saturation to the learning rule.

$$w_t = w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u \quad (4.6)$$

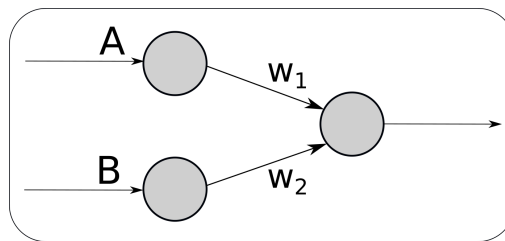
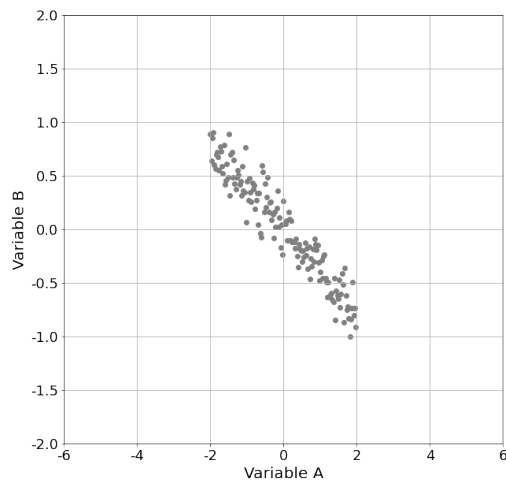
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4.7 Describe the statistical process that takes place when Oja's rule is applied to a dataset.

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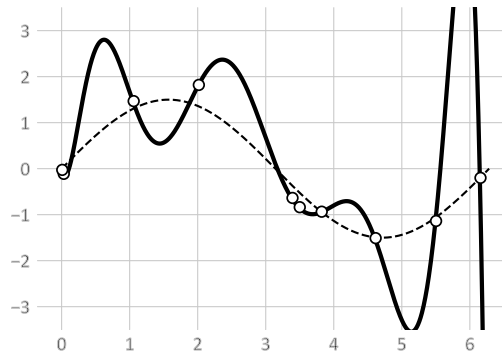
4.8 Consider a system described by two variables A and B. The distribution of these variables is shown in the figure on the left. The variables are used as input of the neural network shown in the figure on the right.



Let Oja's rule compute weights w_1 and w_2 . Give approximate values of w_1 and w_2 .

Problem 5 Machine learning (17 credits)

5.1 In the figure below, a polynomial model (solid line) was fit to a set of noisy data samples (circles) that were recorded from a signal (dashed line). Even though the model has sufficient degrees of freedom, the testing error is too high. What is the name of the issue depicted in the figure? Provide the names of three methods for avoiding it.



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5.2 Name and explain three different learning paradigms.

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5.3 Explain the cross-validation technique and how the performance is measured with the cross validation technique.

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Problem 6 Reinforcement learning (25 credits)

0 ☐ 6.1 Provide a brief explanation of the following three terms/concepts in the context of reinforcement learning as introduced in the lecture: (i) return, (ii) action policy, and (iii) state-value function.

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0 ☐ 6.2 Provide a brief explanation of *Generalized Policy Iteration*. Name and describe the two alternating steps that are performed iteratively.

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The python function below is part of a reinforcement learning agent and is called whenever the action-value function needs to be updated.

```
def train_model(state, action, reward, next_state, next_action):  
    target = reward + gamma * model.predict(next_state)[0][next_action]  
    values = model.predict(state)  
    values[0][action] = target  
    model.fit(state, values, epochs=1)
```

0 ☐ 6.3 State the exact name and category of the reinforcement learning algorithm that the defined function *train_model* implements.

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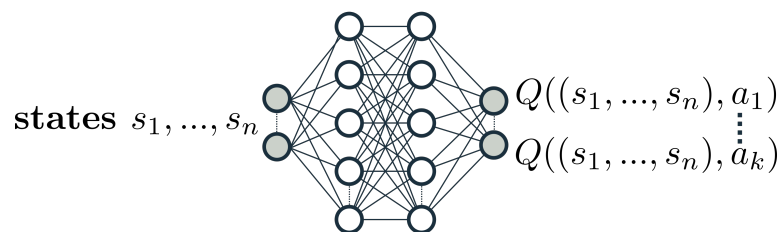
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Architecture 1



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- From the two architectures above, state which is a possible representation for the object *model* based on the context inferred from the code snippet.
- State if the implemented architecture generalizes on states only, actions only, or both.
- State how many times the implemented architecture has to be run in forward direction to find the action that maximizes the action-value function $Q(S_t, A_t)$ at every time step t .

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

